

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.1: Computers and their Components

Concept 3.1.3: Hardware Device Operations

Total Questions: 10

Mark Schemes begin on Page 8

Question 1 | Source: 9618_w25_qp_11 (Q10.a)

10 An architect needs a model of a building.

(a) Explain how the model will be printed using a 3D printer.

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Question 2 | Source: 9618_w25_qp_12 (Q8.b.i)

8 A restaurant accesses software and stores data using cloud computing.

(b) The restaurant uses touchscreens as input devices to allow staff to select items from a menu.

(i) Explain how a touchscreen converts the point of touch to a selection from the menu.

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Question 3 | Source: 9618_s25_qp_11 (Q6.a.i)

6 A programmer is buying a new computer.

- (a) The programmer is considering the following two computers:

Computer 1	Quad-core 2.2 GHz processor 16-bit architecture 1 GB Random Access Memory (RAM) 500 GB magnetic hard disk
Computer 2	Dual-core 3.8 GHz processor 32-bit architecture 2 GB RAM 500 GB solid state drive (SSD)

- (i) Computer 1 has a magnetic hard disk.

Complete the description of the principal operation of a magnetic hard disk by writing the missing words.

The magnetic hard disk has one or more that can be magnetised. These are mounted on a and rotate at high speed.

A is moved across the surface on an arm.

When data is read, the changes in the produce a change in the electric current.

[4]

Question 4 | Source: 9618_w24_qp_11 (Q3.d.i)

- 3 A computer system has a dual-core Central Processing Unit (CPU).

- (d) The computer system is used to store data received from a temperature sensor every five seconds. The data is stored on an optical disc using an optical disc reader/writer.

- (i) Describe the principal operation of an optical disc reader/writer.

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Question 5 | Source: 9618_s24_qp_11 (Q2.c.ii)

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:
- The digital camera in the doorbell starts recording a video.
 - A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell’s internal secondary storage device and overwritten when the secondary storage device is full.



(c) The video doorbell has both primary memory and secondary storage.

(ii) The video doorbell has a solid state (flash) secondary storage device.

Complete the table by writing the answer or answers to each statement about the principal operation of solid state (flash) memory.

Statement	Answer
the two types of logic gate that can be used to create solid state devices	1 2
the number of transistors contained in each cell	
the type of gate that can retain electrons without power	
the type of gate that allows or stops current from passing through	

[4]

Question 6 | Source: 9618_s24_qp_12 (Q2.a)

2 A computer game is being designed that users will be able to play using a virtual reality (VR) headset.

(a) Complete the description of the principal operation of a VR headset.

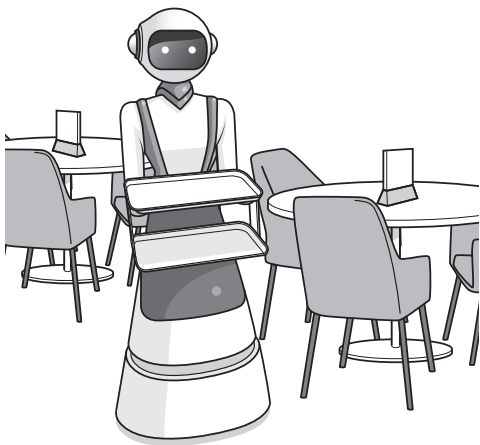
A headset can have one or two that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor.

This sensor is a The data is transmitted to a microprocessor that analyses the data to identify the of movement. Some headsets use that record the user's eye movements for analysis.

[4]

Question 7 | Source: 9618_s24_qp_13 (Q7.d)

7 Robots are used to serve food and drink to customers at a restaurant.



- (d) The robot includes a touchscreen for the customer to make their payment.

Describe the principal operation of a touchscreen.

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Question 8 | Source: 9618_w23_qp_11 (Q7.a)

- 7 (a) Describe the principal operations of a 3D printer.

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Question 9 | Source: 9618_s23_qp_11 (Q4.b)

- 4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

- (b) The central computer stores the video files on secondary storage.

Describe **two** reasons why magnetic storage is more appropriate than solid state storage for this computer.

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Question 10 | Source: 9618_s23_qp_13 (Q3.a)

- 3 A mobile telephone is used to record a video.

- (a) The mobile telephone has a touchscreen. There are different types of touchscreen.

Complete the description of the principal operation of touchscreens.

A touchscreen has two layers. When the user touches the screen, the layers touch and a is completed.

A touchscreen has several layers. When the top layer is touched, there is a in the electric current.

A microprocessor identifies the of the touch.

[5]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q10.a)

10(a)	1 mark per bullet point, max 3 marks Generic mark points: (max 3) - Additive manufacturing - Uses a digital file created from 3D modelling software or Computer Aided Design (CAD) software - The printer builds up the model one layer at a time - The process is repeated for each layer - Some materials may need time to cool and set or UV curing using LEDs for photopolymer resins Specific mark points: (max 1) e.g. - Fused Deposition Modelling (FDM) // Material is heated and pushed through nozzle / extruder - Stereolithography (SLA) // Photosensitive liquid resin is exposed to a UV-laser beam - Digital Light Processing (DLP) // Photosensitive liquid resin is exposed to a UV projected image of the layer to cure the resin - Selective Laser Sintering (SLS) // Uses a laser to form objects from powdered material, the material can be plastics or metals	3
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Mark Scheme for Question 2 | Source: 9618_w25_ms_12 (Q8.b.i)

8(b)(i)	1 mark per bullet point, max 4 marks Methods to determine point of touch (Max 2) - Resistive // two layers of material make contact and complete a circuit - Capacitive // contact with the screen creates a change in charge - Infrared // the beams are broken by the pointing device - Optical imaging // a shadow is created when the surface is touched - Acoustic pulse // the acoustic wave is absorbed by the pointing device - Selection from the menu (Max 2) - The point of touch is used to determine the x and y coordinates - The menu item corresponding to the coordinate position is recognised and the menu item is added to the order	4
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Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q6.a.i)

6(a)(i)	1 mark for each correctly completed space • platters • spindle • read/write head • magnetic field The magnetic hard disk has one or more platters that can be magnetised. These are mounted on a spindle and rotate at high speed. A read/write head is moved across the surface on an arm. When data is read the changes in the magnetic field produce a change in the electric current.	4
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Mark Scheme for Question 4 | Source: 9618_w24_ms_11 (Q3.d.i)

3(d)(i)	1 mark for each bullet point (max 4) • The disc is spun at high speed • A laser is shone onto the disc to read / write • ... using optical head to move it into position • ... it follows the spiral track from the centre outwards • When writing the laser burns pits to represent the data • When reading the laser reflects from pits and lands • The reflection from a pit and a land is different • ... the differences are interpreted as 1 or 0	4
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Mark Scheme for Question 5 | Source: 9618_s24_ms_11 (Q2.c.ii)

2(c)(ii)	1 mark for each row:		4
	Statement	Answer	
	The two types of logic gate that can be used to create solid state devices	NAND NOR	
	The number of transistors contained in each cell	2	
	The type of gate that can retain electrons without power	floating	
	The type of gate that allows or stops current from passing through	control	

Mark Scheme for Question 6 | Source: 9618_s24_ms_12 (Q2.a)

2(a)	1 mark for each correctly completed statement: • (LCD) displays/screens/lenses • gyroscope/accelerometer • direction/speed • digital cameras A headset can have one or two (LCD) displays/screens/lenses that output the image to the user. The headset has speakers that output surround sound to give a realistic experience. The user's head movements are detected using a sensor. This sensor is a gyroscope/accelerometer. The data is transmitted to a microprocessor that analyses the data to identify the direction/speed of movement. Some headsets use digital cameras that record the user's eye movements for analysis.	4
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Mark Scheme for Question 7 | Source: 9618_s24_ms_13 (Q7.d)

7(d)	1 mark each to max 4: • Resistive: The space between the conductive layers is removed/the layers touch and a circuit is completed • Capacitive: The electrical charge changes where the user pressed • The point of contact is identified • ... from the change in electrical field • The software/microprocessor calculates the coordinates	4
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Mark Scheme for Question 8 | Source: 9618_w23_ms_11 (Q7.a)

7(a)	1 mark for each bullet point (max 3) Generic mark points: • Additive manufacturing • Uses a digital 3D model or a Computer Aided Design (CAD) (file) • Builds up the model one layer at a time • ...starting from the bottom • ...using x, y and z co-ordinates • The material is fused / cured together layer by layer Specific mark points: Fused Deposition Modelling (FDM) • Material is heated and pushed through nozzle / extruder Stereolithography (SLA) • Photosensitive liquid resin is exposed to a UV-laser beam Digital Light Processing (DLP) • Uses liquid plastic resin melted with arc lamps Selective Laser Sintering (SLS) • Uses a laser to form objects from powdered material	3
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Mark Scheme for Question 9 | Source: 9618_s23_ms_11 (Q4.b)

4(b)	1 mark for reason, 1mark for application/justification • The computer will have a large number of read/write operations because it is working all the time • .. magnetic storage has more longevity • Magnetic storage costs less per storage unit • ... videos are large files and therefore very large storage capacity is required	4
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Mark Scheme for Question 10 | Source: 9618_s23_ms_13 (Q3.a)

3(a)	1 mark for each term • Resistive • Circuit • Capacitive • Change • Coordinates A resistive touchscreen has two layers. When the user touches the screen, the layers touch and a circuit is completed. A capacitive touchscreen has several layers. When the top layer is touched there is a change/increase/decrease in the electric current. A microprocessor identifies the coordinates of the touch.	5
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